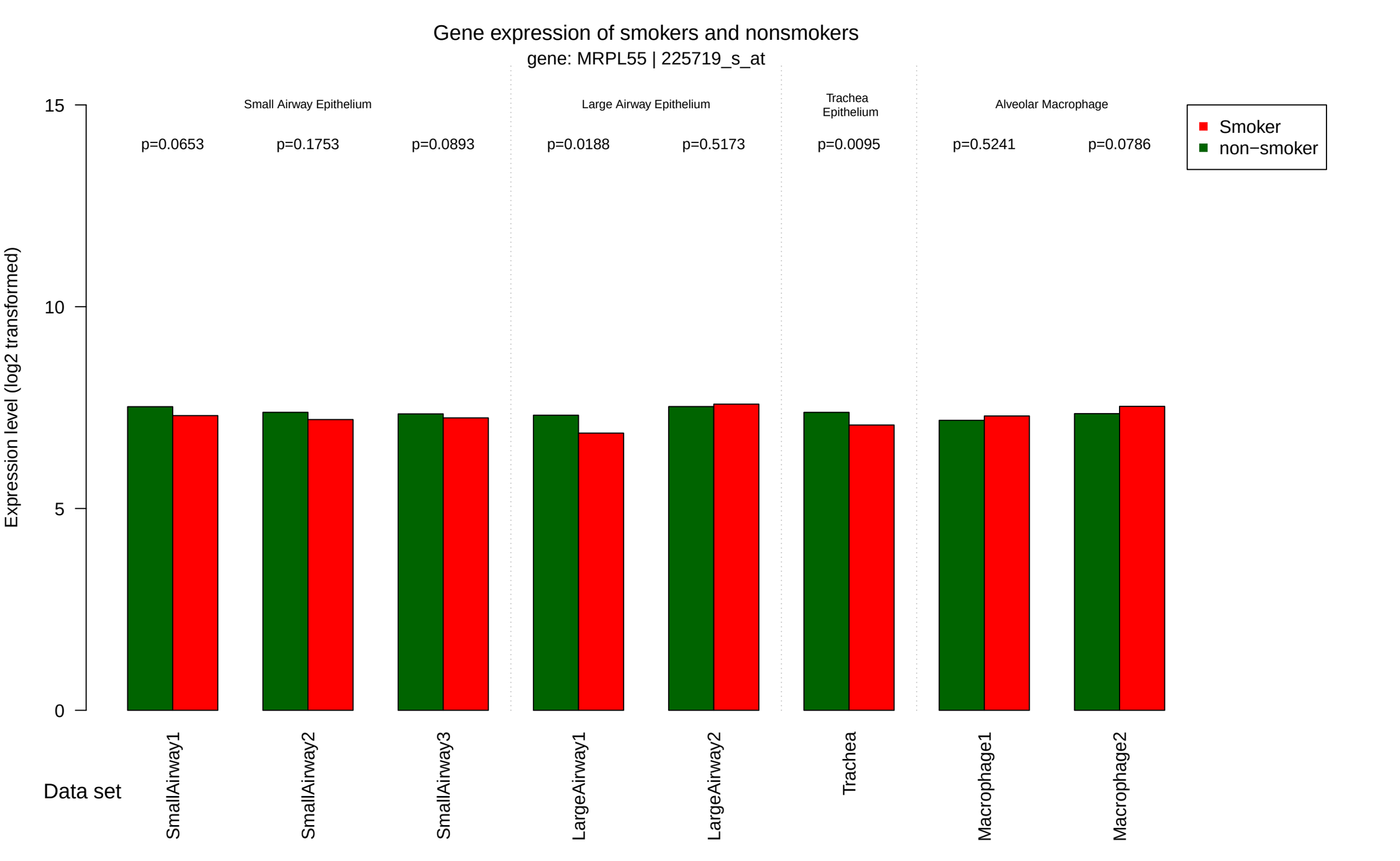
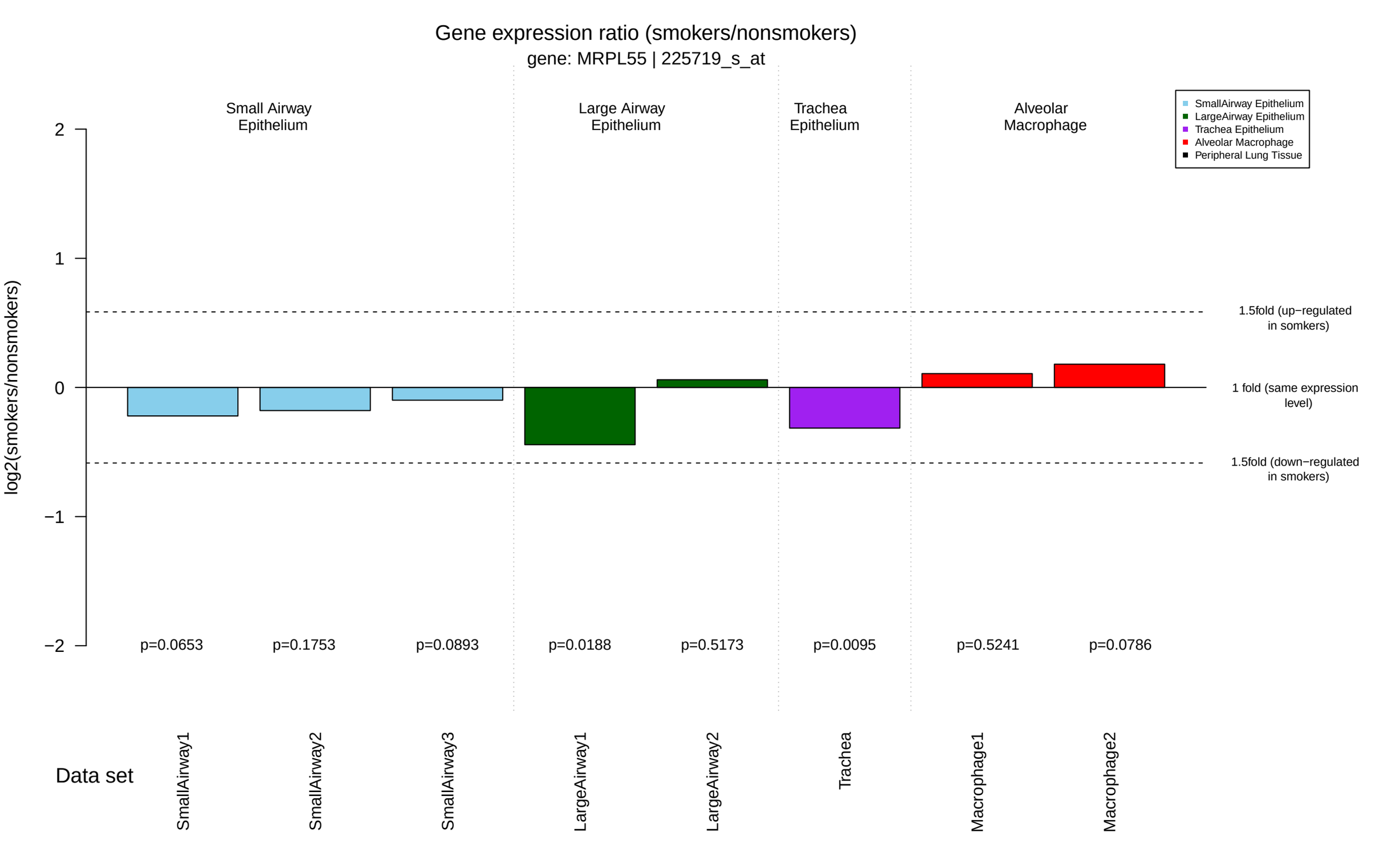
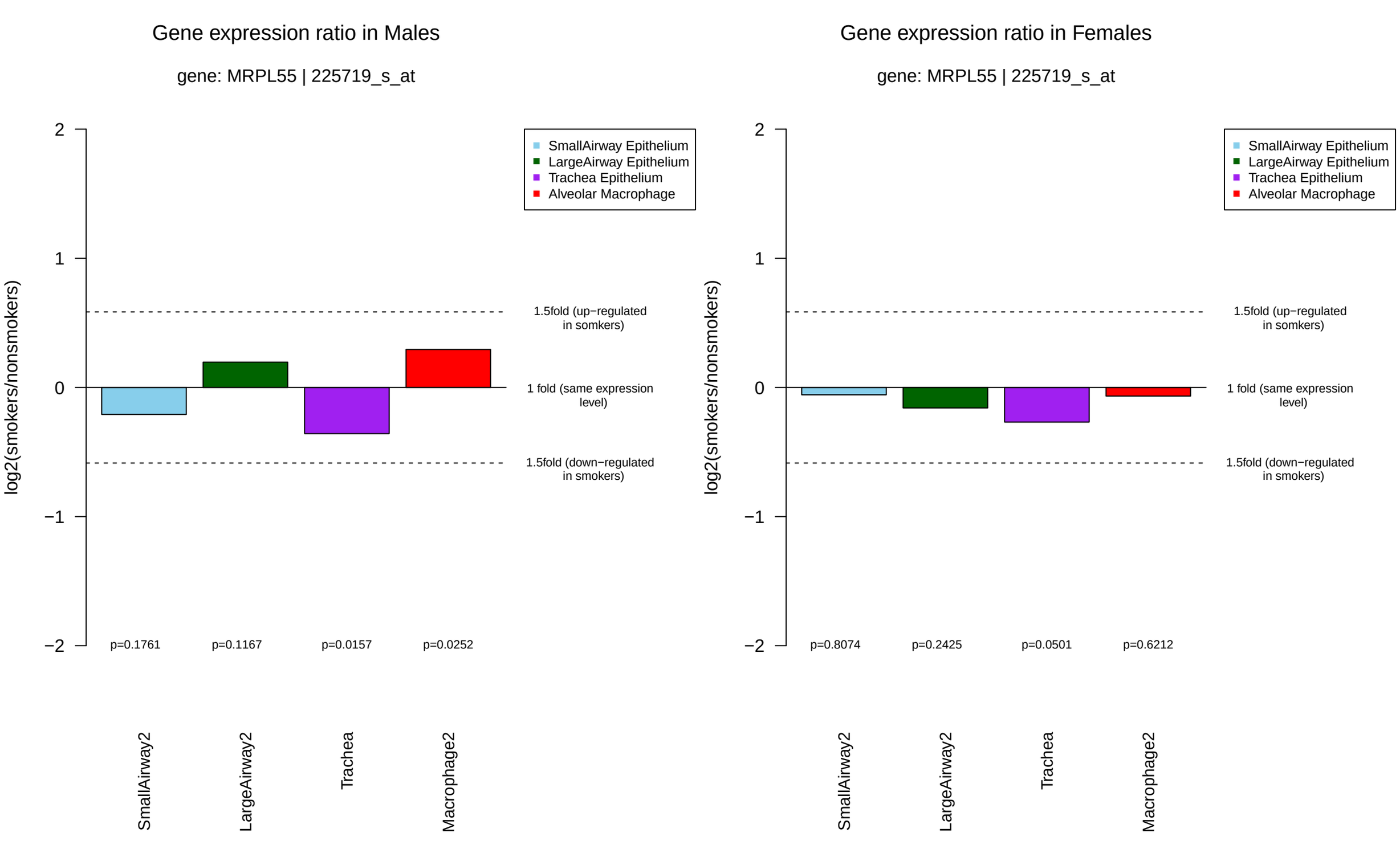
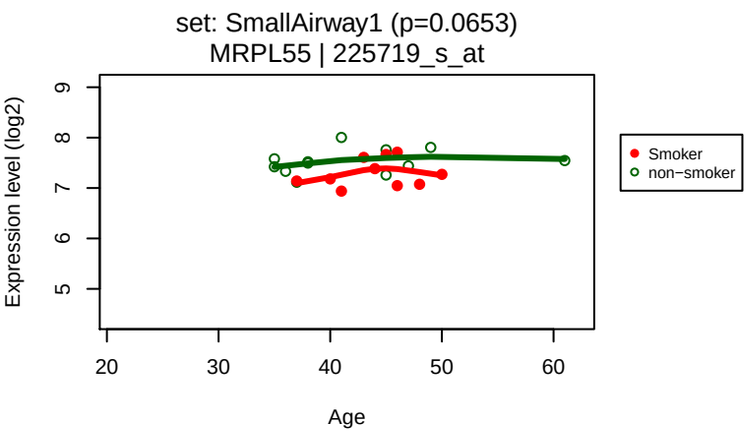
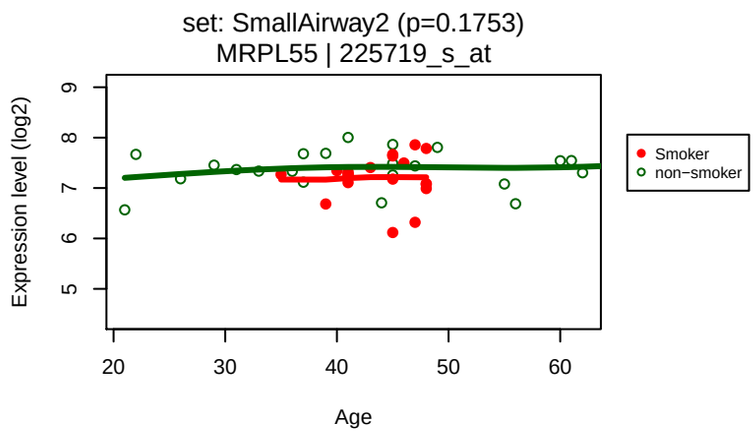
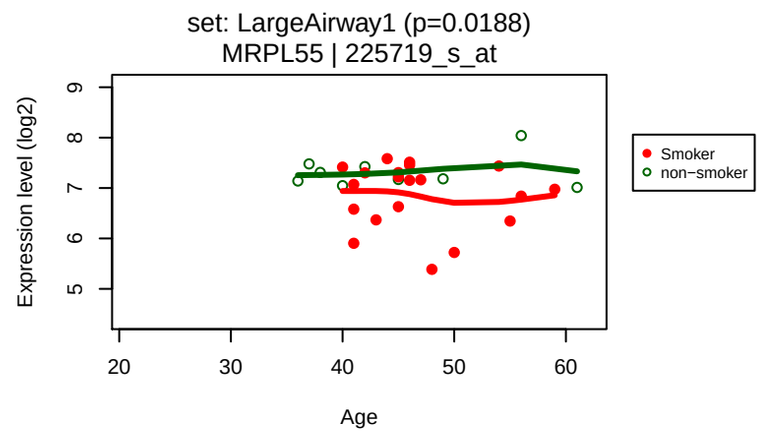

Gene expression of smokers and nonsmokers
gene: MRPL55 | 225719_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0653
p=0.1753
p=0.0893
p=0.0188
p=0.5173
p=0.0095
p=0.5241
p=0.0786
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

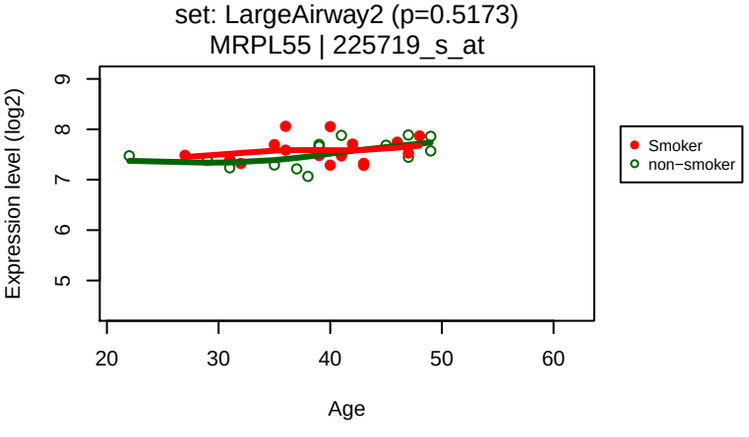

Gene expression ratio (smokers/nonsmokers)
gene: MRPL55 | 225719_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0653
p=0.1753
p=0.0893
p=0.0188
p=0.5173
p=0.0095
p=0.5241
p=0.0786
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

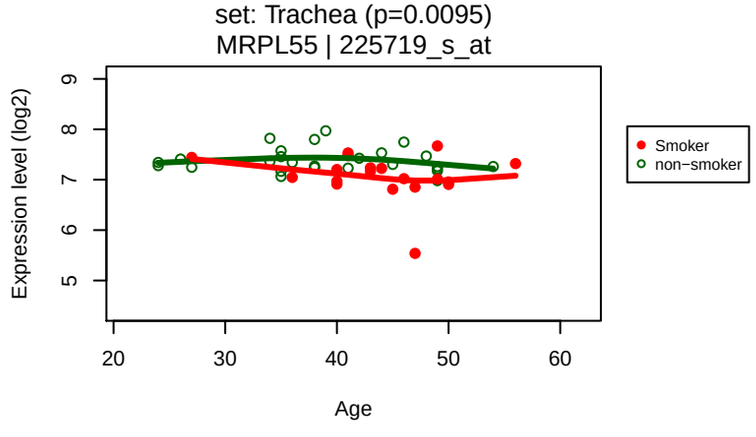

Gene expression ratio in Males
gene: MRPL55 | 225719_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1761
p=0.1167
p=0.0157
p=0.0252
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: MRPL55 | 225719_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8074
p=0.2425
p=0.0501
p=0.6212
SmallAirway2
LargeAirway2
Trachea
Macrophage2

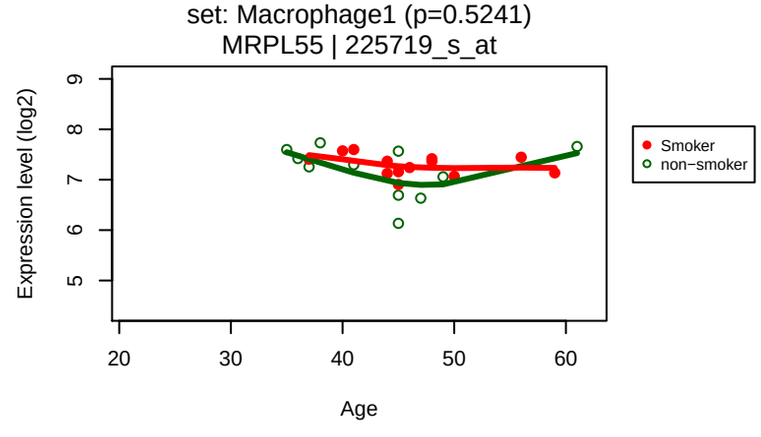

set: SmallAirway1 (p=0.0653)
MRPL55 | 225719_s_at
Expression level (log2)
Age
Smoker
non−smoker

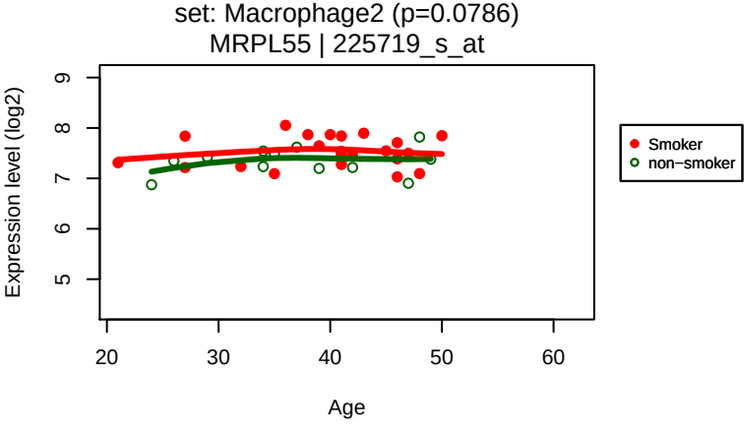

set: SmallAirway2 (p=0.1753)
MRPL55 | 225719_s_at
Expression level (log2)
Age
Smoker
non−smoker

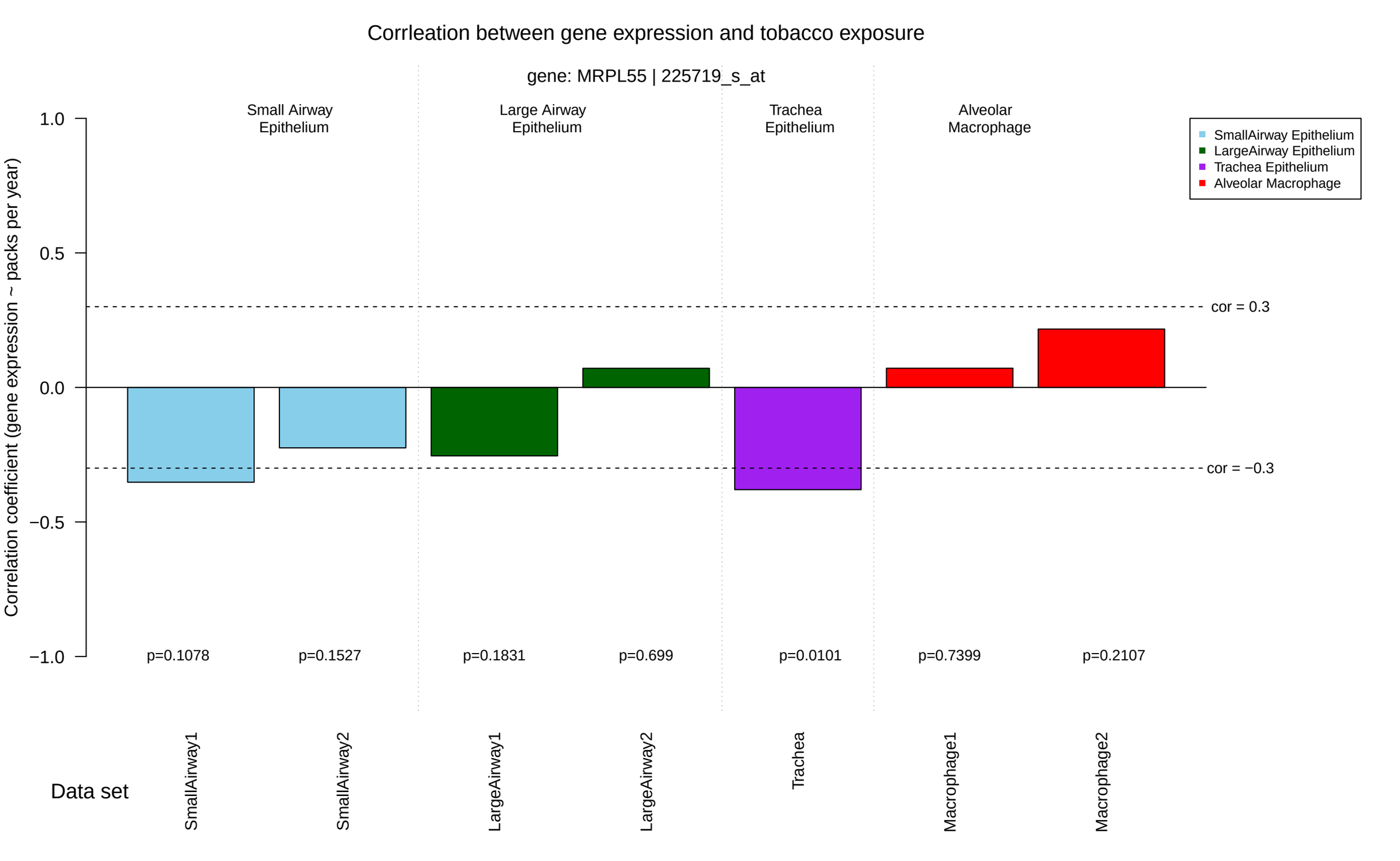

set: LargeAirway1 (p=0.0188)
MRPL55 | 225719_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.5173)
MRPL55 | 225719_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0095)
MRPL55 | 225719_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.5241)
MRPL55 | 225719_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0786)
MRPL55 | 225719_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: MRPL55 | 225719_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1078
p=0.1527
p=0.1831
p=0.699
p=0.0101
p=0.7399
p=0.2107
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2